

Unit 7

ETHICS AND OWNERSHIP

49 items · 236 marks

Syllabus sections in this unit

7.1 Ethics and Ownership

Unit 7: Ethics and Ownership

1. M/J 26/P11/Q2/d

- 2 A computer programmer uses language translators. Two types of translator are an interpreter and a compiler.

(d) The completed computer program is distributed under a licence.

Complete the table by writing the name of each software licence that is described.

software licence	description
	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.
	The software is purchased for a fee. There is no access to the source code.
	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.

[3]

2. O/N 25/P11/Q8/b,c

- 8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(b) Give **three** other ethical considerations for a student using their personal computer to connect to the school network.

1

2

3

[3]

(c) Describe **two** social impacts of students using Artificial Intelligence (AI) to complete their homework.

1

2

[4]

3. O/N 25/P12/Q2/a,b

2 (a) Complete the table by describing each term.

Term	Description
Copyright	
Open Source (Initiative)	
Shareware	
Software Licence	

[4]

(b) Describe the purpose of a code of conduct.

[2]

4. O/N 25/P12/Q10/b

10 A programmer is developing a computer program.

(b) The programmer releases the program as Free Software.

Describe what is meant by Free Software.

[2]

5. O/N 25/P13/Q7/a

- 7 A digital video camera records students in a classroom. The data is transferred from the digital video camera over the internet to a server. Artificial Intelligence (AI) is used to analyse the video on the server to identify when students are interacting with the lesson and the teacher.

(a) Describe **one** ethical impact of this use of AI in the classroom.

[2]

6. O/N 25/P13/Q8/c

- 8 Compilers and interpreters are used to translate programs written in a high-level language into a low-level language.

(c) A programmer is developing an Operating System (OS).

Identify **two** types of software licence the programmer can use to allow other people to edit and redistribute the OS.

1

2

[2]

7. M/J 25/P11/Q4/a

- 4 A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

(a) The new system uses digital cameras and Artificial Intelligence (AI) to identify the customers.

Explain how the new system uses AI to identify each customer.

[4]

8. M/J 25/P12/Q3/a,b,c

- 3** A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

- (a)** Explain how AI is used in the computer program described.

[4]

- (b)** State **two** social benefits of the use of AI in the computer program described.

1

2

[2]

- (c)** The computer program is released under a commercial software licence.

- (i)** Describe the features of a commercial software licence.

[3]

- (ii)** Explain the reasons why an open source software licence might **not** be appropriate for the computer program described.

[3]

9. M/J 25/P13/Q3/d

- 3** A programmer is writing a program in a high-level language.

- (d)** Describe the possible consequences of the programmer **not** joining a professional ethical body.

[3]

10. O/N 24/P11/Q6

- 6** A car park system uses a camera to record the registration number of each car as it enters and leaves the car park.

Explain how artificial intelligence is used in the car park system to identify the car's registration number.

[4]

11. O/N 24/P11/Q7/a,b

7 Software is distributed with a licence.

(a) Give **two** benefits of distributing software using a shareware software licence.

1

2

[2]

(b) Give **two** benefits of distributing software using a commercial software licence.

1

2

[2]

12. O/N 24/P12/Q5/a,b

5 A computer programmer is learning about the importance of acting ethically at work.

(a) Explain the reasons why the computer programmer needs to act ethically towards colleagues and the public.

Colleagues

The public

[4]

(b) The programmer writes a program to be used in a business. The source code of the program can be modified by the business. The business pays the programmer for the maintenance and security updates.

(i) Identify **one** suitable type of software licence for the program **and** describe the reasons for your choice.

Type of software licence

Reasons for choice

[3]

(ii) Explain the reasons why the programmer should copyright their program.

[2]

13. O/N 24/P13/Q5/a,b

5 A computer programmer writes programs that are distributed with a software licence.

(a) Complete the table by writing the type of software licence each statement describes.

Statement	Software licence
A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	
Users can try the software before buying it. Users may not be able to access all features during the trial.	
Software is usually free of charge. Users can access the source code and alter the program to their needs.	

[3]

(b) Explain the reasons why it is important for the computer programmer to join a professional ethical body.

[3]

14. M/J 24/P11/Q5/c(ii)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(c) The bank also needs to keep its customers' data private and secure.

(ii) Customers need to use biometric authentication to access their accounts.
One biometric authentication method is facial recognition.

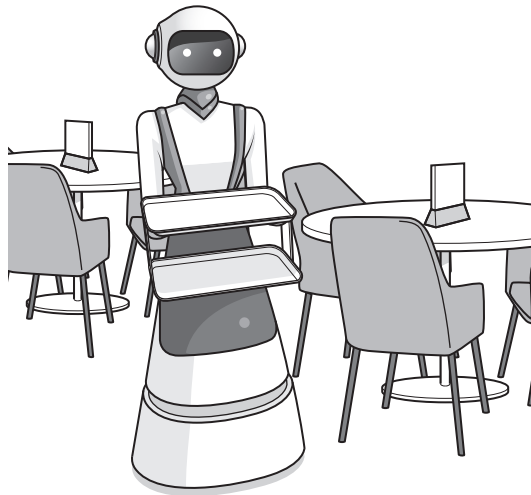
Facial recognition uses Artificial Intelligence (AI).

Describe how AI is used in facial recognition.

[4]

15. M/J 24/P13/Q7/b

- 7 Robots are used to serve food and drink to customers at a restaurant.



- (b) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

[3]

16. O/N 23/P13/Q6/a,b

- 6 (a) State **two** benefits to a programmer of distributing a program using a shareware licence.

1

2

[2]

- (b) Explain why it is important for a programmer to join a professional ethical body.

[4]

17. M/J 23/P11/Q4/c

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (c) The CCTV system uses Artificial Intelligence (AI) to identify the presence of a person in the house and to track their movements.

Describe how AI is used in this system.

[3]

18. M/J 23/P12/Q7/d

7 A software developer is working in a team writing a program for a client.

(d) One section of the program being developed will convert user's speech into commands.

Explain how Artificial Intelligence (AI) can be used in this program.

[3]

19. M/J 23/P13/Q3/b

3 A mobile telephone is used to record a video.

(b) The mobile telephone uses a built-in digital camera to record the video.

The digital camera automatically focuses on the faces of people.

Explain how Artificial Intelligence (AI) is used by the camera to automatically focus on the faces of people.

[3]

20. M/J 23/P13/Q5/b

5 A programmer is developing a computer game in a high-level language to sell to the public.

(b) The programmer needs to publish the game under a software licence so that it can be sold to the public.

Identify the **most appropriate** type of software licence for the game **and** justify your choice.

Licence

Justification

[4]

21. O/N 22/P12/Q9

9 One use of Artificial Intelligence (AI) is for facial recognition software.

Describe the social impact of using facial recognition software to identify individuals in an airport.

[2]

22. O/N 22/P13/Q8/a,b

- 8 (a) (i) Explain why some programs are distributed under an open source licence.

[2]

- (ii) Explain how a programmer benefits from distributing a program under a commercial licence.

[2]

- (b) A commercial program for a vehicle repair garage includes an Artificial Intelligence (AI) module that can diagnose faults and suggest repairs.

Describe **one** economic impact the AI module may have on the garage.

[2]

23. O/N 21/P11/Q4/a,b(i),c

- 4 Francis is starting his first job as a software developer for a multinational company.

- (a) Francis has been advised to join a professional ethical body.

Describe the benefits to Francis of joining a professional ethical body.

[3]

- (b) Francis is shown the software he will be working on. He is unfamiliar with the Integrated Development Environment (IDE) he is required to use.

- (i) Describe the ways in which Francis can act ethically in this situation.

[2]

- (c) Francis is part of a team writing a program. He finds an error in part of the program that has already been tested. He decides not to tell anyone because he is worried about the consequences.

Explain the reasons why Francis acted unethically in this situation.

[2]

24. O/N 21/P13/Q4/a,b(i),c

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(c) Francis is part of a team writing a program. He finds an error in part of the program that has already been tested. He decides not to tell anyone because he is worried about the consequences.

Explain the reasons why Francis acted unethically in this situation.

[2]

25. M/J 21/P11/Q1/d

1 Anya scans an image into her computer for a school project.

(d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

26. M/J 21/P12/Q2/a,b,c

2 Aisha manages a team of software developers.

- (a) Explain the reasons why it is important that Aisha acts ethically in relation to her team members.

[2]

- (b) The team are developing a computer game where the user plays a board game (such as chess) against the computer.

Describe how the computer would use Artificial Intelligence (AI) to play the board game.

[3]

- (c) The final game will be released under a licence.

Tick (✓) **one or more** boxes in each row to identify the licence(s) each statement describes.

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code				
The user must always pay before being able to use the software				
The user can redistribute the software				
The user always gets a trial period				

[4]

27. M/J 21/P13/Q1/d

1 Anya scans an image into her computer for a school project.

- (d) Anya made sure that the image was not subject to any copyright before scanning it.

Describe what is meant by **copyright**.

[2]

28. 9608 M/J 21/P11/Q1

- 1 Draw **one** line from each software licence to its correct description.

Software licence**Description**

Shareware

A limited version of the software could be released and downloaded by anyone, but users would need to pay to unlock additional features.

Open
Source

A licence **must** be purchased to use the software.

Commercial

Users **cannot** download the software over the Internet.

The original source code is made available for other developers who can then modify and improve the software.

[3]

29. 9608 M/J 21/P11/Q7/a,b,c

7 This question presents three scenarios.

Tick (✓) **one** box for each scenario to indicate whether you think the behaviour shown is ethical or unethical. Justify your choice.

- (a) Marina has a very short deadline to create a piece of software for a client. Algorithm A is quick to code and simpler to test but has an inefficient run time. Algorithm B is more complex to code and test but has a more efficient run time. She decides to implement algorithm A.

Ethical	
Unethical	

Justification

[2]

- (b) Doug is managing a project for a client that has fallen behind schedule. He asks all members of the project team to work extra hours and to cancel any holiday plans to get the project back on schedule.

Ethical	
Unethical	

Justification

[2]

- (c) Debbie is programming a car safety management system. She thinks that the test plan she has been given is not adequate. She decides to discuss her concerns on a public internet forum.

Ethical	
Unethical	

Justification

[2]

30. 9608 M/J 21/P12/Q6/a,b,c

6 This question presents three scenarios.

Tick (✓) **one** box for each scenario to indicate whether you think the named person's behaviour is ethical or unethical. Justify your choice.

- (a) Latifah has changed jobs and has started to work for a new company. The company uses an Integrated Development Environment (IDE) to develop code. Latifah decides not to use the IDE that the company has because she is familiar with a different IDE.

Ethical	
Unethical	

Justification

[2]

- (b) Samid is in charge of a project to write a banking application. He is employing staff to work on the application. His daughter is a computer security expert. She is looking for a new job. Samid decides to employ his daughter.

Ethical	
Unethical	

Justification

[2]

- (c) Jason works for a social media company. He is concerned that users of the company's social media website have not been told how their personal data is being used.

Jason tells his manager his concerns. His manager tells him not to worry because there have been no complaints from the users. Jason takes no further action.

Ethical	
Unethical	

Justification

[2]

31. 9608 M/J 21/P13/Q7/b,c

- 7 Patrick is writing a new software application. He is using a compiler to develop the software application.

- (b) Patrick has completed the application. He needs to choose whether to distribute the software application using an open source licence or a commercial licence.

Describe **open source** and **commercial** software licensing.

Open source

Commercial

[4]

- (c) Patrick works for a company that has a code of conduct for its employees.

Explain the reasons why the company needs a professional code of conduct.

[3]

32. M/J 20/P12/Q4/a,b

- 4 Sophie is about to start a new job as a junior software developer.

- (a) She is worried about joining a new team of people.

- (i) State **one ethical** action that Sophie can take to help her to feel more confident about starting work.

[1]

- (ii) State **two ethical** actions that Sophie's manager can take to help Sophie to feel more confident about starting work.

1

2

[2]

- (iii) State **one ethical** action that Sophie's new colleagues can take to help Sophie to feel more confident about starting work.

[1]

- (b) Explain why Sophie is asked to sign a professional code of conduct before starting work.

[3]

33. M/J 20/P13/Q4/c

4 Annchi is writing a computer game with a group of friends.

(c) Annchi needs to decide which type of software licence to use for the game.

(i) Give **two** benefits to **Annchi** of using a commercial licence.

1

2

[2]

(ii) Give **one** benefit to the **customers** of the game being released using a commercial licence.

[1]

(iii) Describe **one** benefit to the **customers** of the game being released using a shareware licence.

[2]

34. O/N 19/P12/Q5/c

5 Mica has created some software and has copyrighted it. She wants to stop other people from copying and changing it illegally.

(c) Mica is releasing the software under a commercial licence.

(i) Give **two** benefits to Mica of using a commercial licence.

1

2

[2]

(ii) Name **two** other types of software licence.

1

2

[2]

35. M/J 19/P11/Q2/a

- 2** A software company produces software and distributes it under different software licences.

(a) Four descriptions of software licences are given.

Write the type of software licence that best fits each description. Use a different type of licence for each description.

1. The software can be legally used, only after a fee has been paid.

Licence type

2. The source code comes with the software. If the software is modified, the edited source code must be released under the same conditions as the original software.

Licence type

3. The software is free for a trial period and then a fee is requested, or expected, if the user wants to continue to use the software.

Licence type

4. The source code comes with the software. The software is free to be downloaded, edited, and distributed, possibly without restriction.

Licence type

[4]

36. M/J 19/P12/Q4/d

- 4** Shazia is creating a computer program that will be released to the public. The program includes a video.

(d) Shazia wants to make sure her computer program is copyrighted.

(i) Define the term **copyright**.

[1]

(ii) Shazia does not want to release the software as open source.

Explain why Shazia does **not** want to use an open source licence.

[2]

(iii) Name **and** describe **two** software licences, other than open source that Shazia could use.

Licence 1

Licence 2

[2]

37. M/J 19/P13/Q2/d

- 2 Frankie is a software developer. He is developing a program to manage customer records for a client with an online retail business. He must ensure that data stored about each customer are both secure and private.

(d) One principle of the ACM/IEEE Software Engineering Code of Ethics is to always act in the best interest of the client.

Explain how Frankie can ensure that he is acting in the best interest of his client.

[3]

38. O/N 18/P11/Q3/a,b

- 3 Hugo has produced a program (app) for mobile phones. He needs to decide whether to use an Open Source licence or to distribute the app as shareware.

(a) Describe what is meant by **Open Source licence** and **shareware**.

Open Source

Shareware

[4]

(b) Tick (✓) **one** box to indicate the licence Hugo should use. Justify your choice.

Open Source	☐
Shareware	☐

Justification

[2]

39. O/N 18/P11/Q5/a,b,c

- 5** For each of the following scenarios, tick (✓) **one** box for each scenario to indicate whether you think the person's behaviour is ethical or unethical. Justify your choice.

- (a)** Kevin is a software engineer who has recently started a job with a new company. He is using program code from his previous employer in his new employer's programs.

Ethical	
Unethical	

Justification

[2]

- (b)** Nadya is a software developer. She has accepted a new job. She has never worked with the programming languages used by this new company. Nadya is planning to increase her knowledge of these programming languages before she starts her new job.

Ethical	
Unethical	

Justification

[2]

- (c)** Maria finds that one of her team members has produced some inventive code. She presents this to her manager, stating that it was produced by the team. She does not mention the individual's name.

Ethical	
Unethical	

Justification

[2]

40. O/N 18/P12/Q5/a,b,c

- 5** This question presents three scenarios. For each scenario, tick (✓) **one** box to show whether you think the person's behaviour is ethical or unethical. Justify your choice.

- (a)** Wendy is a software engineer who is developing a program for her company. Her friend, Noah, is developing a program for a different company. Wendy looks at the code that Noah is writing to get ideas for her own program.

Ethical	
Unethical	

Justification

[2]

- (b)** Amit is fixing some bugs in the computer system of a large multinational company. He is asked to sign a confidentiality agreement. He sees some confidential information which contains the names of other multinational companies that have broken the law. He copies this information and releases it on the Internet.

Ethical	
Unethical	

Justification

[2]

- (c)** Farah is providing a company with an estimate for the cost of writing a program. The company she works for is in financial difficulty so she increases the estimate by 10%.

Ethical	
Unethical	

Justification

[2]

41. O/N 18/P13/Q3/a,b,c

- 3** This question presents three scenarios. Tick (✓) **one** box for each scenario to indicate whether you think the person's behaviour is ethical or unethical. Justify your choice.

- (a)** Mason is using his work computer to book a holiday whilst at work.

Ethical	
Unethical	

Justification

[2]

- (b)** Ethan is supervising a trainee. The trainee asks Ethan for a reference for another job. Ethan does not want to lose the trainee, so refuses to give him a reference.

Ethical	
Unethical	

Justification

[2]

- (c)** Margarita finds that one of her team members has produced some inventive code. She presents this to her manager, praising the individual by name.

Ethical	
Unethical	

Justification

[2]

42. O/N 18/P13/Q5/a

- 5** Arnold is a software developer. He has created a computer game for people to download over the Internet. Arnold is considering releasing the game as a piece of commercial software.

- (a) (i)** Describe what is meant by a **commercial licence**.

[2]

- (ii)** Name and describe **one** other type of licence that Arnold can consider using.

Licence type

Description

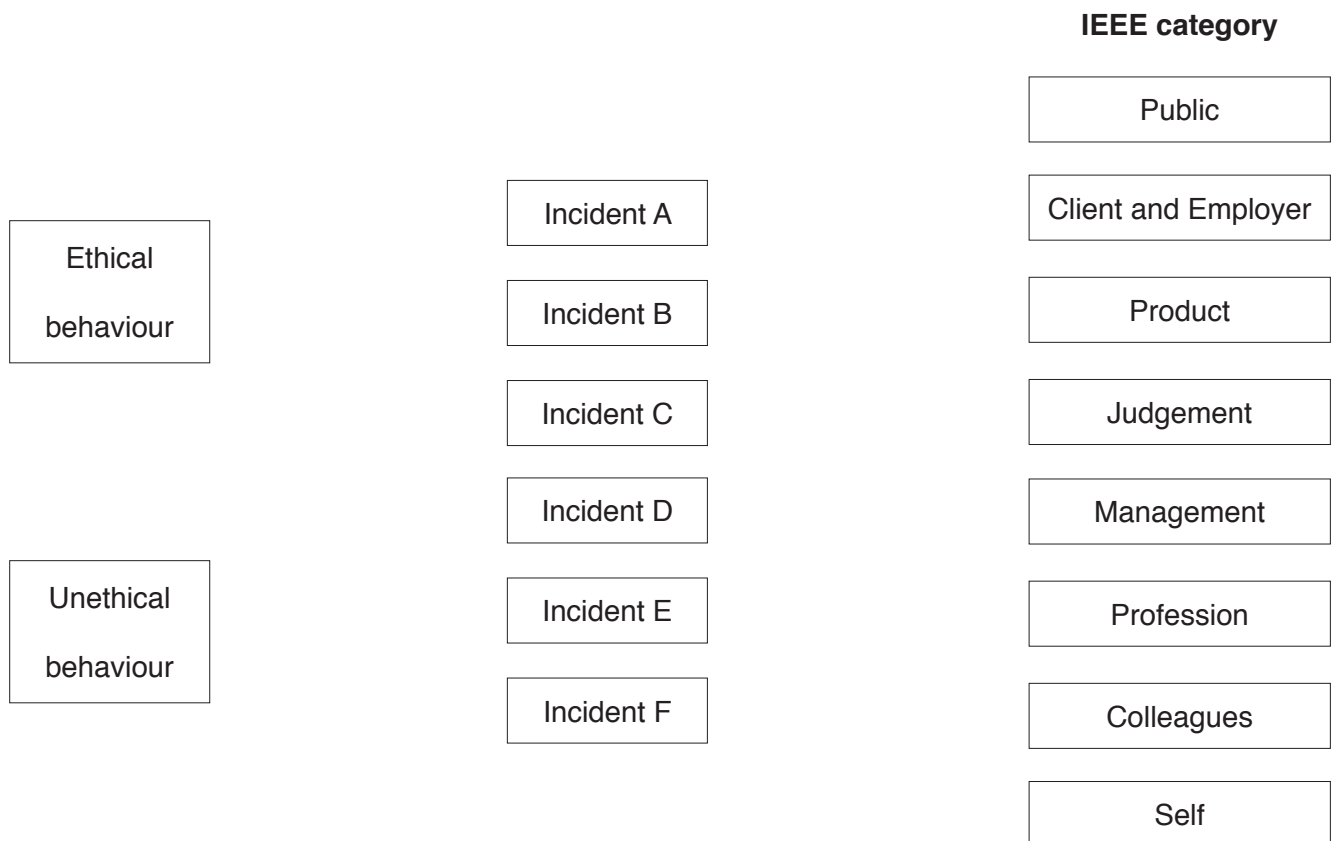
[3]

43. O/N 17/P12/Q5/a

- 5 The IEEE Software Engineering Code of Ethics uses eight key principles shown in the right-hand column of the following diagram.

Tom is employed as a tester with a software company. He is keen to become a trainee programmer.

The middle column in the diagram labels six incidents which have happened to Tom this week. The table that follows the diagram describes each incident.



Incident	Description
A	Tom has received some phishing emails. He reported this to the bank they were supposed to have come from.
B	Tom has asked his manager if they will pay for him to attend a programming course.
C	Tom is testing beta versions of new games software at work. He copies the software on to CD-Rs and sells them to his friends.
D	Tom has completed the application forms to join the Chartered Institute for IT.
E	Tom finds it difficult to work with one of his colleagues. His way of dealing with this has been to refuse to speak with the colleague.
F	Tom's manager had considered the testing of a new game was completed. Tom reported to his manager that he thought there were still bugs which needed to be rectified.

- (a) Draw a line on the diagram to link each of the six incidents to either ethical behaviour or unethical behaviour. [2]

44. O/N 17/P13/Q6/a

- 6 Raj has joined a software company as a trainee programmer. He was given the company's Code of Conduct document during his induction training. The handbook has a section headed 'Ethical Behaviour'.

(a) Describe what is meant by **ethics**.

[2]

45. M/J 16/P11/Q5

- 5 Three types of software licensing and four descriptions are shown in the table below.

Put a tick (✓) in each row to match each description to the appropriate type of software licensing.

Description	Type of software		
	Open source	Shareware	Commercial
Software is purchased before it can be used			
Source code comes with the software			
Software is provided free on a trial basis			
The software can be modified by the user			

[4]

46. M/J 16/P12/Q5

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The software can be modified by the user			

[4]

47. M/J 16/P13/Q3/a,b,c

3 A company needs new software to manage its accounts. It is evaluating two different options. One option is open source software and the other is commercial software.

(a) Explain what is meant by open source software.

[2]

(b) Explain what is meant by commercial software.

[2]

(c) The company has decided to purchase commercial software.

Identify **four** benefits to the company in choosing the commercial software option.

1

2

3

4

[4]

48. O/N 15/P11/Q4

- 4 Paul works part-time for a large software company. The company sells security software to a number of banks. He also runs his own software company that produces and sells computer games.

Six statements about computer ethics are shown below.

Draw lines to indicate whether each statement describes ethical or unethical behaviour.

Statement

To save time, Paul fakes the test results when testing the bank security software.

Paul uses the software developed in his day job to help write some of the games software routines.

To allow him to concentrate on his games software, Paul has frequently turned down job opportunities in his day job.

To make the games software more realistic, Paul uses password protection code used in the bank security software.

Because his work load is increasing, Paul is now using overseas companies to write some of the routines used in his games software.

Paul carries out training on how to write games software in his spare time.

Ethical

Unethical

49. O/N 15/P13/Q4

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Six statements about computer ethics are shown below.

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To allow him to concentrate on his games software, Paul has frequently turned down job opportunities in his day job.

To make the games software more realistic, Paul uses password protection code used in the bank security software.

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Paul carries out training on how to write games software in his spare time.

Ethical

Unethical

Answers Section

1. M/J 26/P11/Q2/d

2(d)	1 mark for each correct licence		3 Do not accept copyleft for open source Do not accept freeware for open source
	Software licence	Description	
	Shareware	The software is released on a trial period. After the trial period ends, a fee may be required to continue using the software.	
	Commercial // proprietary	The software is purchased for a fee. There is no access to the source code.	
	Open source initiative // free software	The software is released with the source code. The software can be changed and distributed with the changes under the same licence restrictions.	

2. O/N 25/P11/Q8/b,c

8(b)	1 mark per bullet point, max 3 marks e. g. - Students should consider privacy issues - Students should not make inappropriate use of social networking // Students should not engage in cyberbullying - Students should not copy work that is not their own - Students should abide by the school rules/guidelines for accessing the network - Students should only use the school network for legitimate purposes - Students should not hack / crack other computers	3
8(c)	1 mark per bullet point, max 4 marks e. g. Negative Impacts - If students use AI to shortcut learning, they may miss out on developing reasoning and problem-solving skills // Students may not be learning anything // It could lead to a decline in specific skills - It could have negative impacts on student communication // heavy reliance on AI may limit opportunities for collaboration, teamwork, and face-to-face communication - Emphasises the digital divide// some students may not have access to the technology - The AI answer might not be correct leading to mis-information - Increased screen time and isolation could contribute to anxiety or loneliness Positive Impacts - Students given additional support may do better - Pupils who struggle in traditional settings may gain self-esteem through AI-assisted learning at their own pace Long-Term Impacts - Protection of intellectual property // ethical issues and considerations - Schools may need to redesign curricula and/or assessment to integrate AI	4

3. O/N 25/P12/Q2/a,b

2(a)	1 mark for each correct row, max 4 marks	4										
<table><tr><th>Term</th><th>Description</th></tr><tr><td>Copyright</td><td>Gives the holder the legal right of ownership of intellectual property</td></tr><tr><td>Open Source (Initiative)</td><td>Software can be freely copied, distributed or adapted</td></tr><tr><td>Shareware</td><td>Gives the user a trial period / limited functionality before a payment/sign up may be required</td></tr><tr><td>Software Licence</td><td>A (legal) agreement that defines the rights and terms of usage for the software</td></tr></table>			Term	Description	Copyright	Gives the holder the legal right of ownership of intellectual property	Open Source (Initiative)	Software can be freely copied, distributed or adapted	Shareware	Gives the user a trial period / limited functionality before a payment/sign up may be required	Software Licence	A (legal) agreement that defines the rights and terms of usage for the software
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2(b)	1 mark per bullet point, max 2 marks e.g. • to create a safe, respectable and professional working environment for all employees • to ensure that employees understand the expectations • ... and the consequences of their actions • to provide a clear outline of what is considered good practice • so there are clear rules to follow in certain situations • to protect the reputation of the organisation // maintain public trust	2										

4. O/N 25/P12/Q10/b

10(b)	1 mark per bullet point, max 2 • The software is free of restrictions • The software is not necessarily free of charge • The <u>source code</u> comes with the program // A user can edit the <u>source code</u> to add functionality / suit their needs • Users can share the software with others	2
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5. O/N 25/P13/Q7/a

7(a)	1 mark per point, max 2 marks e.g. • The learning experience could be improved • ... by identifying students who are struggling • and employing more personalised learning • There could be early Intervention • ... to determine which students require extra support • Teachers and students could understand the challenges better • ...and identify optimal times for learning • There could be privacy concerns • students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • due to constant monitoring and pressure • contributing to student stress and anxiety	2
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6. O/N 25/P13/Q8/c

8(c)	1 mark per bullet point, max 2 marks • Open Source (Initiative) • Free Software (Foundation)	2
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7. M/J 25/P11/Q4/a

4(a)	1 mark each to max 4 e.g. • Uses image recognition / facial recognition • Measures the distance between facial features of the customer • and stores the values in a database of user information • An image is captured using the digital camera • the AI identifies that the image is a face • by analysing the pixels to find patterns • the distance between features is calculated and compared to those in database	4
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8. M/J 25/P12/Q3/a,b,c

3(a)	1 mark each to max 4 e.g. • Uses image recognition // optical character recognition • to analyse a photograph/pixels to identify the location of characters within an image • in order to convert the pattern of pixels into individual characters • Uses natural language • to combine the characters into words • and compare the words to database to identify a match in the language of the writing • it may also predict what is missing / incomplete / from what has been read • then compare the words to a database of the user's chosen language to find a match and translate • Uses text-to-speech • to generate the audio waveforms corresponding to the words	4
3(b)	1 mark each to max 2 e.g. • It can help people with visual impairment to identify which products to purchase • It can help people who cannot understand the language • It can help to overcome learning / reading difficulties	2
3(c)(i)	1 mark each to max 3 • The program is usually sold for a fee • Users have no access to source code • It protects the intellectual rights of the programmer / developer • Users cannot legally alter or share the program • There are limitations on the use of the software e.g. number of times it can be installed • Support and maintenance is usually provided e.g. updates, bug fixes etc.	3
3(c)(ii)	1 mark each to max 3 e.g. • The source code would be accessible therefore it could be changed • which might lead to the program outputting an incorrect response • so people could purchase the wrong item • which might have serious consequences // by example e.g. allergies	3

9. M/J 25/P13/Q3/d

3(d)	1 mark each to max 3 e.g. • There is no / limited access to legal advice • in case action is taken against them • There are fewer networking opportunities • so they could miss out on contacts / jobs • There would be less access to training • There would be no clear laid out ethical guidelines • and / or people to discuss potential ethical problems with • possibly leading to inappropriate / unethical actions • which might lead to legal proceedings / recourse	3
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10. O/N 24/P11/Q6

6	1 mark for each bullet point (max 4) e.g. • It uses image recognition • The pixels of each image from the camera are stored • ... and matched to the expected shape/size/colour of registration number • It uses optical character recognition • The pixels within the registration number area identified are analysed • ... and compared to expected characters/letters/numbers	4
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11. O/N 24/P11/Q7/a,b

7(a)	1 mark for each bullet point (max 2) • The user can test/trial the program before purchasing to see whether it meets needs • There is no initial cost • Users who try the software may spread the word so the developer gets more custom • The developer can get feedback to improve the program	2
7(b)	1 mark for each bullet point (max 2) • The owner/developer can earn a fee • Comprehensive support is usually offered so problems can be easily solved • The software is usually well tested so few bugs	2

12. O/N 24/P12/Q5/a,b

5(a)	1 mark for each bullet point max 2 for public max 2 for colleagues e.g. Colleagues: The programmer should: • Treat colleagues fairly and avoid acts of discrimination • Be prepared to accept / offer critique • ... so that technical work can be improved • Credit the contributions of colleagues • ... so that they feel valued / respected • Help / train colleagues etc. e.g. The public: The programmer should: • Maintain the health, safety and welfare of the public • ... so as not to endanger anyone • Be honest / realistic in making claims about the software • ... so that public trust is maintained • Maintain the security of the public's data etc.	4
5(b)(i)	Type of software licence 1 mark for • Open Source Initiative // Free Software Foundation Reason for choice: 1 mark for each bullet point (max 2) • Allows program source code to be examined • ... so that the business can adapt/modify the program to meet their needs • Allows the programmer to earn money for the software and updates	3
5(b)(ii)	1 mark for each bullet point (max 2) • To identify themselves as the owner/author // To gain formal recognition of ownership • To allow for legal consequences if anyone steals/copies it • To restrict competition	2

13. O/N 24/P13/Q5/a,b

5(a)	1 mark for each correct licence		3
	Statement	Software Licence	
	A fee is charged for the software. The source code cannot be accessed. Users do not get a free trial.	Commercial	
	Users can try the software before buying it. Users may not be able to access all features during the trial.	Shareware	
	Software is usually free of charge. Users can access the source code and alter the program to their needs.	Open Source // Free software	
5(b)	1 mark for each bullet point (max 3) e.g. • The professional ethical body has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so the computer programmer does not have to decide what is ethical it's written down • Clients/staff know the computer programmer is reputable • ... there may be a test/requirement for entry • ... giving recognition of the computer programmer's skills/knowledge • The professional ethical body provides help and support • ... for example, if the computer programmer needs legal advice • The professional ethical body runs training courses • ... to keep the computer programmer's skills up to date		3

14. M/J 24/P11/Q5/c(ii)

5(c)(ii)	1 mark each to max 4 : e.g. • Captures an image of the face • Uses image recognition • Trained to identify the features of a face in an image • ... using a large number of images • Analyse images for facial features • Uses the probability of a match	4
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15. M/J 24/P13/Q7/b

7(b)	1 mark each to max 3 : e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • ... programmed action(s) for matching sentence(s) are performed	3
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16. O/N 23/P13/Q6/a,b

6(a)	1 mark for each bullet point (max 2). • Users are able to trial the program and may offer the programmer constructive feedback • More people might try the program because the trial is free of charge • Allows bugs to be found and corrected on a wide range of computer system configurations improving the programmer's original program • Users that find the trial useful will buy the program so programmer gets income • Allows the program to be copyrighted and so protects the programmer's intellectual property rights // no illegal modification allowed	2
6(b)	1 mark for each bullet point (max 4). • The programmer has ethical guidelines to follow • ...so purchasers of programs know the standards being applied • ...so programmer does not have to decide what is ethical or not • Clients know programmer is reputable • ...recognition of programmer's skills/knowledge • There may be an entry requirement / exam • ... so clients know that the programmer is competent • Professional ethical body provides help and support • Such as legal advice // by appropriate example • Enables programmer to attend the ethical body's training courses • ...to keep skills up to date // to develop skills	4

17. M/J 23/P11/Q4/c

4(c)	1 mark each to max 3 Examples: • Uses image recognition • Monitors every image taken to identify matching images/shapes/features to a 'person' ... • ... starts recording to secondary storage/permanently when a person is identified • System identifies direction of movement of person and uses this to decide where/how to move the camera/record • System identifies other cameras to start recording based on direction of movement	3
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18. M/J 23/P12/Q7/d

7(d)	1 mark each to max 3 • Uses speech recognition • ... which identifies key phrases / words spoken • ... and matches these to a database • ... and generates the most likely sentence / command / word	3
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19. M/J 23/P13/Q3/b

3(b)	1 mark each to max 3 Examples: • Scans the scene in real time • Identifies if there are faces in the image • Uses facial recognition • ... uses image recognition • ... takes each frame individually • ... analyses the pixels • ... stores pattern for a face • ... looks for patterns that match/come close to the pattern for a face • Camera focuses on the pattern identified	3
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20. M/J 23/P13/Q5/b

5(b)	1 mark for appropriate licence; 1 mark for each point to max 3 • Commercial software licence • User has to pay for the product so the programmer can gain an income • Enables the program to be copyrighted • ... so the user cannot legally edit the program // the programmer retains control over product • ... and can take legal action against people who attempt to illegally copy it /sell it on • Shareware licence • Enables the program to be copyrighted • The user cannot legally edit the program so the developer retains control over product • User can try the program for free and then pay for the full game which allows the programmer to gain an income • so more people can experience it and therefore be more likely to buy it	4
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21. O/N 22/P12/Q9

9	1 mark for each bullet point (max 2). For example: • incorrect recognition of faces leads to mistakes such as • ... access to facilities / systems may be denied • privacy issues / people do not like data being stored • individuals will feel safer • ... there might be a reduction in crime • faster boarding • catching criminals	2
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22. O/N 22/P13/Q8/a,b

8(a)(i)	1 mark for each bullet point (max 2): • to allow users to customise the code • to allow errors to be reported / identified / fixed by users • to allow additional features to be added to the code • to allow for collaboration	2
8(a)(ii)	1 mark for each correct point (max 2) Example: • enables the program to be copyrighted • prevents illegal changes to the program / protects the source code • prevents illegal copies of the program being made • a fee can be charged for the program	2

8(b)	1 mark for a correct economic impact and 1 mark for corresponding description Example: • reduce costs to the garage • ... because less time taken for diagnosis • increase profits for the garage • ... as technicians spend more time repairing, so completing more jobs in a day • decrease costs passed to customer • ... so garage may gain customers • profit margins can be reduced • ... because program may be expensive to buy / maintain / update	2
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23. O/N 21/P11/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

4(c)	1 mark per bullet point to max 2 e.g. • He didn't act in best interest of product • ... because the product might fail because he didn't report the error • He didn't act in best interest of client • ... because if the product does not work then they have been let down because he didn't report the error • He didn't act in the best interest of the profession • ... he is letting his profession down because he didn't report the error • He didn't act in best interest of the company • ... not correcting the error early could lead to later problems	2
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24. O/N 21/P13/Q4/a,b(i),c

4(a)	1 mark per bullet point to max 3 e.g. • He has ethical guidelines to follow • ... so clients/other staff know the standards being applied • ... so he does not have to decide what is ethical it's written down • Clients / staff know he is reputable • ... recognition of his skills / knowledge • ... there may be a test / requirements for entry • They provide help and support • ... for example if he needs legal advice • They run training courses • ... to keep his skills up-to-date	3
4(b)(i)	1 mark per bullet point to max 2 e.g. • He can tell the manager he has not used it • ... and how he will get up-to-date • He can perform his own research on how to use it • He can explain to the manager that he needs additional training • He can(ask the manager to book on a training course • He can ask for a mentor / to shadow someone • He can practice at home before starting	2

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25. M/J 21/P11/Q1/d

1(d)	1 mark per bullet point to max 2 • The formal and legal rights to ownership // intellectual property rights • Protects against unauthorised reproduction of work • Provides for legal right of redress	2
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26. M/J 21/P12/Q2/a,b,c

2(a)	1 mark per bullet point to max 2 • To make sure the team members feel valued • To get the best work out of the team • To enable them to work well together • To enable them to create the best product for the client	2
2(b)	1 mark per bullet point to max 3 • The rules / past moves / decision making algorithms of the game will be stored • The AI program is trained, by playing many times • AI will look (ahead) at possible moves • ... and/or analyse the pattern of past choices • ... and choose the move most likely to be successful • Computer could learn how to improve // learn from previous mistakes • ... by storing the positive/negative result of choices • ... and changing its future choices	3

2(c)

1 mark for each correct column

4

Statement	Free Software Foundation	Open Source Initiative	Shareware	Commercial Software
The user can edit the source code	✓	✓		
The user must always pay before being able to use the software				✓
The user can redistribute the software	✓	✓	✓	
The user always gets a trial period			✓	

27. M/J 21/P13/Q1/d

1(d)	1 mark per bullet point to max 2 - The formal and legal rights to ownership // intellectual property rights - Protects against unauthorised reproduction of work - Provides for legal right of redress	2
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28. 9608 M/J 21/P11/Q1

1	1 mark for each licence correctly identified <table><thead><tr><th>Software licence</th><th>Description</th></tr></thead><tbody><tr><td>Shareware</td><td>A limited version of the game could be released and downloaded by anybody, but users would need to pay to unlock additional features</td></tr><tr><td>Open Source</td><td>A licence must be purchased to use the software</td></tr><tr><td>Commercial</td><td>The software cannot be downloaded from the Internet</td></tr><tr><td></td><td>The original source code is made available for other developers who can then modify and improve the software</td></tr></tbody></table>	Software licence	Description	Shareware	A limited version of the game could be released and downloaded by anybody, but users would need to pay to unlock additional features	Open Source	A licence must be purchased to use the software	Commercial	The software cannot be downloaded from the Internet		The original source code is made available for other developers who can then modify and improve the software	3
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29. 9608 M/J 21/P11/Q7/a,b,c

7(a)	1 mark per bullet point for each justification, to max 2 Either Unethical • Marina knowingly produces a solution that is not as efficient as it could be • ...and this does not enhance the reputation of the person/company • Reference to IEEE standards <u>in context</u> Or Ethical • Marina acts in the best interests of the employer to complete a working product on time • ...that is more likely to be correct as the coding is simpler • Reference to IEEE standards <u>in context</u>	2
7(b)	1 mark per bullet point for each justification, to max 2 Either Unethical • Doug has a management responsibility • ...to be fair to his colleagues • ...but is showing a lack of personal integrity by applying pressure • Reference to IEEE standards <u>in context</u> Or Ethical • Doug is trying to act in the best interests of his employer/client • ...and may be asking employees to voluntarily offer to work additional hours • ...he may be paying them extra • Reference to IEEE standards <u>in context</u>	2
7(c)	1 mark per bullet point for each justification, to max 2 Either Unethical • Debbie may be breaking company policy discussing current projects // confidentiality • ...and has a duty to act in the best interests of his company • ...and should have raised her concerns internally • Reference to IEEE standards <u>in context</u> Or Ethical • Debbie may feel that in the public interest the problem should be discussed • ...to ensure that a safety critical system is fully tested • ...so as to prevent potential loss of life • Reference to IEEE standards <u>in context</u>	2

30. 9608 M/J 21/P12/Q6/a,b,c

6(a)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical • Latifah believes that her productivity is better using an IDE that she has experience of • ... and the work is still compatible with other IDEs • ...the IDE is open source so does not cost anything extra • Reference to IEEE standards <u>in context</u> Or Unethical • Latifah is knowingly using software that is not provided or supported by the company • ...and this may lead to issues with future software compatibility/maintenance • Reference to IEEE standards <u>in context</u>	2
6(b)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical • Samid believes his daughter is qualified and would be the best person for the job • ...and believes that it is in the best interest of his client to secure a high-quality employee as soon as possible • ... they may have interviewed her with a third party and she came out as the best candidate • Reference to IEEE standards <u>in context</u> Or Unethical • Samid shows poor judgement by favouring his daughter • ...rather than using a fair and open recruitment process • there may be people with better qualifications that are being overlooked • Reference to IEEE standards <u>in context</u>	2
6(c)	1 mark per bullet point for each justification, to max 2 e.g. Either Ethical • Jason has already raised his concerns (to his manager) • ...and has fulfilled his professional responsibilities by doing so. • Reference to IEEE standards <u>in context</u> Or Unethical • Jason should continue to raise any concerns he has with a higher level of management in the company • ...because it may be against the public interest for users to have their data used in this way without them explicitly knowing it. • Reference to IEEE standards <u>in context</u>	2

31. 9608 M/J 21/P13/Q7/b,c

7(b)	1 mark per bullet point, max 2 marks per licence Open Source • The (source) <u>code</u> is distributed with the software • users can modify the software • The software is (usually) free of cost • Users must redistribute the modified software under the same terms Commercial • The software is (usually) purchased for a cost • A licence defines how it can be used • The source code is not distributed // users cannot change the software	4
7(c)	1 mark per bullet point to max 3 e.g. • To identify what employees can do • To identify what employees cannot do • To identify the repercussions of employees performing activities they should not • To identify the company's/employee's responsibilities • To identify the company's values/missions • To identify what behaviour/events/activities can be measured against • To identify how they will comply with legislation	3

32. M/J 20/P12/Q4/a,b

4(a)(i)	1 mark only e.g. • Read about the languages she will be using • Visits the office prior to starting • Speaks to her manager about concerns	1
4(a)(ii)	1 mark per action to max 2 For example: • Prepare an induction • Invite Sophie in before starting • Introduce Sophie to the team • Give Sophie a mentor	2
4(a)(iii)	1 mark only e.g. • Prepare a greeting/introduction • Provide structured support • Invite Sophie to social event(s) before/at the start to meet people	1

4(b)	1 mark per bullet point to max 3 • Sophie is confirming that she understands the code of conduct • To make sure Sophie knows what behaviour is expected of her • To make sure Sophie understands there may be consequences of some actions • To ensure all employees adhere to the same standards	3
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33. M/J 20/P13/Q4/c

4(c)(i)	1 mark per benefit to max 2 • She can charge a fee for the game • She retains the copyright • ... so, the game cannot be re-distributed by a third-party without her permission	2
4(c)(ii)	1 mark per benefit to max 1 • Potentially better support, as she is charging a fee • Likely to have fewer bugs / less prone to malware than if distributed under other licences e.g. open source • Redress available if the game does not function correctly	1
4(c)(iii)	1 mark per bullet point • They can check it works // check if it meets their requirements • ... without having to paying a fee if it does not	2

34. O/N 19/P12/Q5/c

5(c)(i)	1 mark per benefit to max 2 • So that she can sell the software for a fee // make money from the software • A commercial licence prohibits unauthorised/further copies being made and/or distributed • A commercial licence prohibits any changes to the software	2
5(c)(ii)	1 mark per bullet point to max 2 • Open Source • Free Software • Shareware • Freeware	2

35. M/J 19/P11/Q2/a

2(a)	1 mark for each correct term • Commercial Licence • Free Software Licence • Shareware Licence • Open Source Licence	4
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36. M/J 19/P12/Q4/d

4(d)(i)	1 mark for correct answer Formal or legal recognition of ownership of the program // Formal or legal restriction / permissions on use of the program // The intellectual property rights to the program	1
4(d)(ii)	1 mark per bullet point • She does not wish to release the source <u>code</u> • She does not want anyone to be able to edit / modify / share the source code/program • She wants to make money from the program	2
4(d)(iii)	1 mark for a <u>name and a description</u> of each licence to max 2 Commercial Software • The program is purchased for a fee • It restricts the number of users/possible time period for use // Limited number of installations allowed // Software key needed to install • Source code not provided // source code protected / cannot be edited • Anyone can purchase/download if agree to the terms Shareware • The program is free for a trial period // The (free) program may have limited functionality // Need to purchase / enter details after trial • Users do not have access to the source code // source code may not be edited • Users may re-distribute the software. Freeware • There is no charge for the software • The software could still be copyrighted • She can set her own restrictions on what a user can do with the program	2

37. M/J 19/P13/Q2/d

2(d)	1 mark per bullet point to max 3 For example: He should ... • ...Keep the client's personal data private • ...Involve the client in the development // ...Communicate with the client • ...Provide the solution that the client asked for • ...Keep the project running on time // budget • ...Keep the client informed of any problems/delays	3
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38. O/N 18/P11/Q3/a,b

3(a)	1 mark per bullet point, max 2 marks per licence Open Source • The source code is released with program • Users can edit the source code to suit their needs • Users re-release their version under the same terms • Can be cost-free but may also need payment Shareware • Users get a free trial of the software • ... which may be limited in features • No access to source code // Program cannot be edited • Then they have to pay / sign-up after the expiry date // Then they have to pay / sign-up to get full functionality // Then they have to pay / sign-up to stop unwanted pop-ups, etc.	4
3(b)	1 mark per bullet point to max 2 marks for chosen licence Open Source For example: • Hugo does not have to set up ways to take funds • Others may enhance / improve the program • Hugo can charge a fee for the App <u>Or</u> Shareware For example: • Hugo can charge for the App • Not giving away the code/people can't copy the code • ... Hugo gets the sole recognition for the program • Possible legal consequences if someone does copy the code • If users need to sign up, their data can be used for marketing etc. • Customers have peace of mind that the software hasn't been maliciously edited / bugs introduced	2

39. O/N 18/P11/Q5/a,b,c

5(a)	1 mark per bullet point for each justification, to max 2 Either Unethical • Work belongs to the company it was created for // copyright • ... Kevin cannot use it without permission • It reduces the integrity of the person / profession / new company • Reference to IEEE standards <u>in context</u> Or Ethical • The program code could be open source • Kevin might own the copyright of code • Kevin may have permission to use the code • Reference to IEEE standards <u>in context</u>	2
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5(b)	1 mark per bullet point for each justification, to max 2 Either Unethical • Nadya has accepted a role / work she knows she cannot do • This reduces the integrity of the person • She may let down the new organisation who are expecting her to be able to do the work • Reference to IEEE standards <u>in context</u> Or Ethical • She is taking steps to prepare for the role • ... Without expecting the company to do it • Nadya may have told the company that she didn't know the languages but that she would learn them • Reference to IEEE standards <u>in context</u>	2
5(c)	1 mark per bullet point for each justification, to max 2 Either Ethical • The individual works as part of the team ... • ... therefore, the team should / will get the credit • Maria is not lying about who produced it • Reference to IEEE standards <u>in context</u> Or Unethical • Maria should identify who / where the idea originated • It prevents the individual getting recognition • Maria is not being supportive of the individual • Reference to IEEE standards <u>in context</u>	2

40. O/N 18/P12/Q5/a,b,c

5(a)	1 mark per bullet point for each justification, to max 2 Either Unethical • Noah's work may be confidential • Wendy shouldn't claim someone else's ideas / work as her own • She is bringing the profession into disrepute • Reference to IEEE standards <u>in context</u> Or Ethical • The code could be open source • Wendy may have permission from Noah • Wendy isn't copying the code, just getting ideas • Reference to IEEE standards <u>in context</u>	2
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5(b)	1 mark per bullet point for each justification, to max 2 Either Unethical • Amit has a responsibility to his company • He should have taken it to the police rather than putting it on the Internet • He has a signed agreement to say he will not give anything away • Reference to IEEE standards <u>in context</u> Or Ethical • Amit is acting in the public interest • Amit may not have actually signed the confidentiality agreement • If acting illegally, the multinational company should be brought to justice • Reference to IEEE standards <u>in context</u>	2
5(c)	1 mark per bullet point for each justification, to max 2 Either Ethical • It might save people's jobs • Farah is acting in the best interest of her company • Reference to IEEE standards <u>in context</u> Or Unethical • Farah has a responsibility to act in the best interest of her client • It could give her company a bad reputation • Reference to IEEE standards <u>in context</u>	2

41. O/N 18/P13/Q3/a,b,c

3(a)	Either Ethical • He is booking the holiday in his own time / lunchtime // he is self-employed • He has been given permission • Reference to IEEE <u>in context</u> Or Unethical • Should not use company computer for personal use • Should be working whilst at work • Reference to IEEE <u>in context</u>	2
3(b)	Unethical: Max 2 marks from • Company will get a bad reputation • Should be supporting his colleague • Reference to IEEE <u>in context</u>	2

3(c)	Either Ethical • She is supporting her colleague • Working in the best interests of the company • Reference to IEEE <u>in context</u> Or Unethical • Praising one team member instead of the whole team • Others in the team may also have contributed, so she is not being supportive of the whole team • Reference to IEEE <u>in context</u>	2
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42. O/N 18/P13/Q5/a

5(a)(i)	1 mark per bullet point, max 2 • Restricted use • Source code not provided // source code protected • Anyone can purchase/download if agree to the terms • Limited number of installations allowed // Software key needed to install	2
5(a)(ii)	1 mark for name and 2 marks for description Either Open Source • The source code is released with the program • Users can change / edit the source code to enhance the game • Users can re-release the game under the same terms // the game might spread more easily Or Shareware • Users get free trial or limited access for set time • Users do not have access to the source code // source code may not be edited • At end of trial period, users may have to pay or register to continue using the game // Can get people 'hooked' and then charge a fee Or Freeware • There is no fee for the game • The game could be copyrighted • Modification, re-distribution or reverse engineering of the game without permission is prohibited	3

43. O/N 17/P12/Q5/a

5(a)	Mark as follows: Unethical: C and E 1 Mark Ethical: A,B, D and F 1 Mark	2
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44. O/N 17/P13/Q6/a

6(a)	Two marks from: • A system of moral principles • That guide behaviour / decision making • Based on philosophical / religious views • By example, e.g. respectful and considerate behaviour	Max 2
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45. M/J 16/P11/Q5**5 One mark** per row.**[4]****No mark** if more than **one tick** in any row.

Description	Open source	Shareware	Commercial
Software is purchased before it can be used			✓
Source code comes with the software	✓		
Software is provided free on a trial basis		✓	
The software can be modified by the user	✓		

46. M/J 16/P12/Q5**5 One mark** per row.**[4]****No mark** if more than **one tick** in any row.

Description	Open source	Shareware	Commercial
Software is purchased before it can be used			✓
Source code comes with the software	✓		
Software is provided free on a trial basis		✓	
The software can be modified by the user	✓		

47. M/J 16/P13/Q3/a,b,c**3 (a) Two** from:**[2]**

- The source code comes with the software.
- The user can edit the source code.
- Once edited, the software is re-distributed with the changes.

(b) Two from:**[2]**

- The software is purchased.
- With a **licence** which restricts the number of users / possible time period for use.
- The program code for the software cannot be edited.

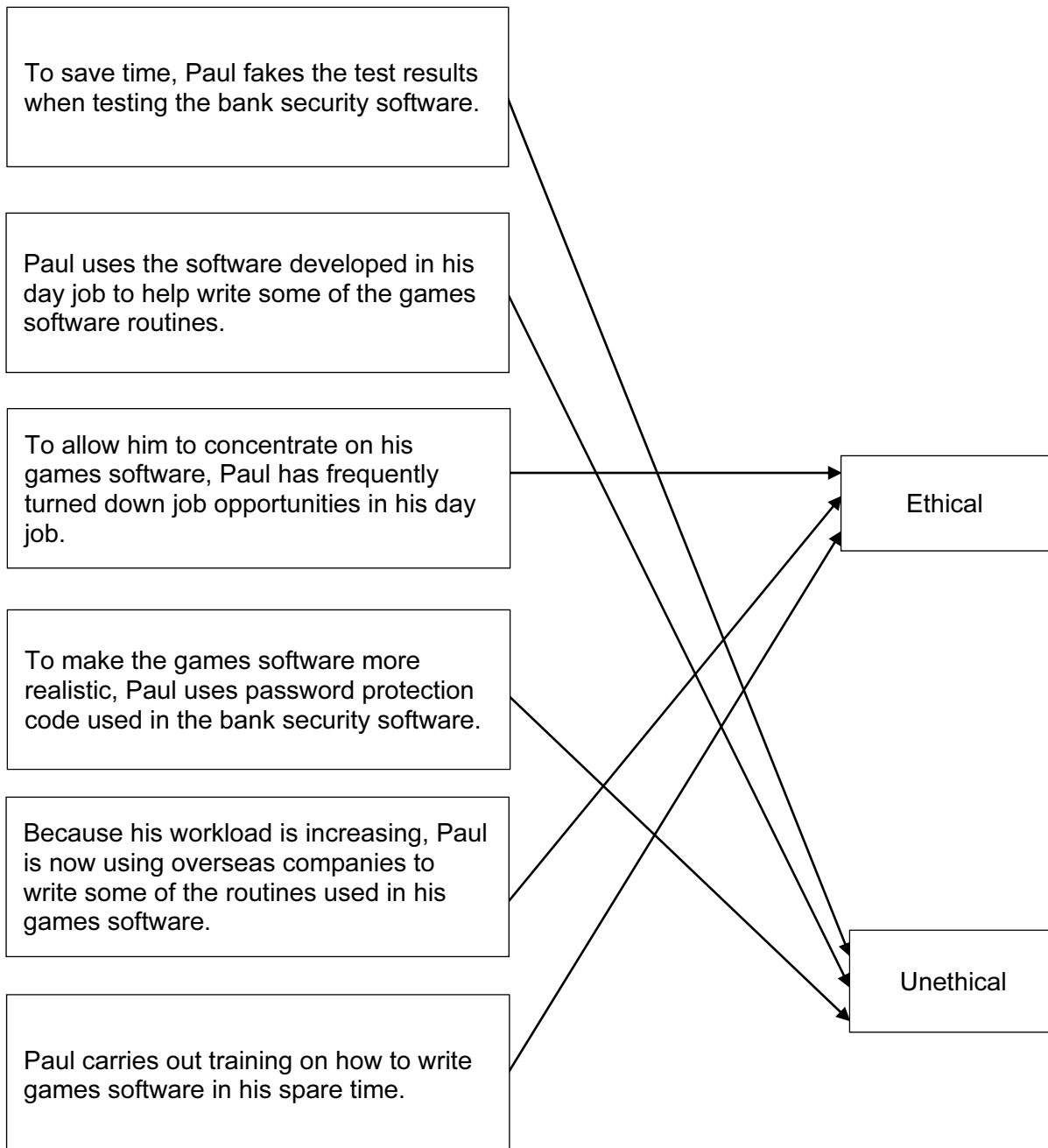
(c) Four from:

[4]

- Support / training is readily available so help can be accessed if needed.
- More robust software / fewer bugs as it has been tested more thoroughly/by more users.
- Forums / user groups will exist for popular software.
- Software upgrade path likely to be available (at minimal cost).
- Manufacturer develops patches that can be automatically downloaded.
- Compatibility is inbuilt for other commercial software.

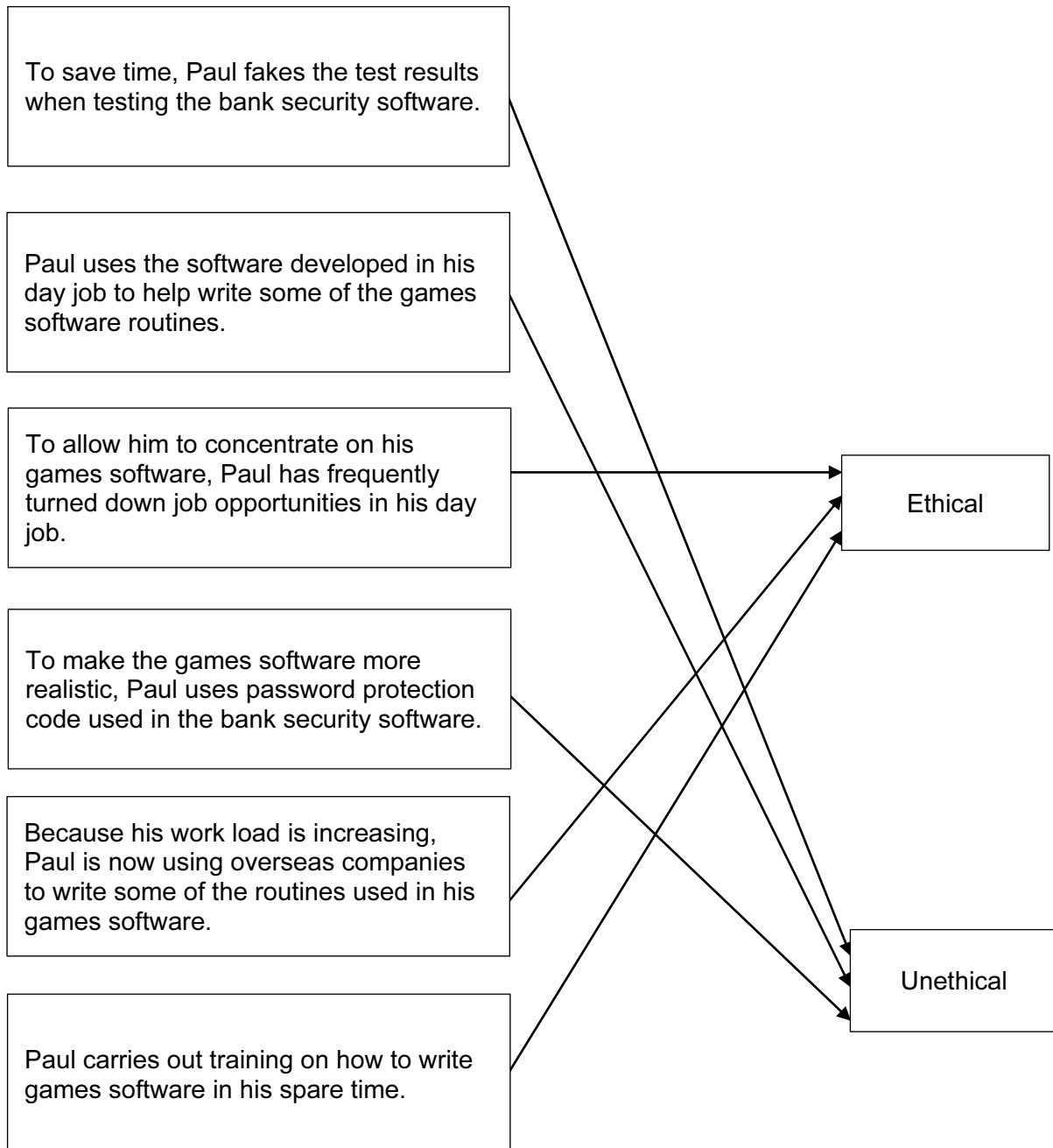
48. O/N 15/P11/Q4

4



1 mark for each correct line, two lines from one box is incorrect

[6]

49. O/N 15/P13/Q4**4**

1 mark for each correct line, two lines from one box is incorrect

[6]